1. Acyclic closures.

Acyclic closures are not exclusive to resolving k (rather than $R = Q/I$), but they're helpful to constructing a minimal free resolution of k that works in any case (while $Q[X]$ min. model is a minimal algebra).

① Def: an acyclic closure over R, $R\langle Y\rangle$, is a semifree extension of R that resolves k, where we adjoin exterior in the odd, and the derived power variables in the even degree.

Derived power:

$$x^{(i)} \cdot x^{(j)} = \binom{i+j}{i} x^{(i+j)}, \qquad \underbrace{\partial(x^{(i+1)}) = \partial(x)x^{(i)}}_{\text{directly defined by this } \partial}.$$

- Compared to $\partial(x^n) = n\partial(x)x^{n-1}$, no annoying coeff. around
- In cases where k has char. 0, acyclic closure & min. model agree.
- Prop.: $\partial(x^{(i+1)}) = \partial(x)x^{(i)}$ does satisfies the Leibniz's rule.

Pf: Use $\binom{n-1}{k} + \binom{n-1}{k-1} = \binom{n}{k}$.

② Properties (very similar to $Q[X]$).

i) $\alpha: A \to B$ map of dg alg. (Surj., local, or $\simeq$)

$\Rightarrow \exists!\ \tilde{\alpha}: A\langle X : \partial(x_z) = z\rangle \longrightarrow B\langle X : \partial(x_z) = \alpha(z)\rangle$

(Surj., local, or $\simeq$).

ii) Thm: Acyclic closures $R\langle Y\rangle$ and $R\langle Z\rangle$ of k are isomorphic

dga's. They induce $k\langle Y\rangle \xrightarrow{\sim} k\langle Z\rangle$.

iii) Thm: $R\langle Y\rangle \simeq \hat{R}\langle Y\rangle$ (Note that they're over R & $\hat{R}$).

Pf:

- Using finite length complex $M \xrightarrow{\sim} M\otimes\hat{R}$, we have:

$$R\langle Y_{\le n}\rangle \longrightarrow R\langle Y_{\le n}\rangle\otimes\hat{R}.$$

- By i), $R\langle Y_{\le n}\rangle\otimes\hat{R} \xrightarrow{\sim} \hat{R}\langle Y_{\le n}\rangle$ by induction, since $R\langle Y_{\le n}\rangle\otimes R$ is indeed adjoining Y_n to $R\langle Y_{\le n-1}\rangle$ such that $\partial(y_z)=z\otimes 1$, and $z\otimes 1 \mapsto \hat{z}$.

Cor: $\varepsilon_n(R)=\varepsilon_n(\hat{R})$.

2. Derivations.

Def: the <u>derivations</u> $\varepsilon_i(R) := |Y_i|$, $R\langle Y\rangle$ acyclic closure of k.

e.g.

$\varepsilon_1(R)=\mu(m)=\text{embdim}(R)$, $\varepsilon_2(R)=\mu(H_1(R\langle Y_1\rangle))=\mu(H_1(K^R))=\mu(I)$

since $R\langle Y_1\rangle$ is exactly $Kos(\underline{m})$.

① Thm (A-S): An acyclic closure for k is a minimal free res. for k.

pf: See Thm 3.3.1 in BEJ or Thm 6.3.4 in AVR.

use Serre's lemma.

② Calculating $\varepsilon_n(R)$.

i) We know as res. of k:

$$\cdots \longrightarrow \begin{matrix} RY_3 \\ \oplus \\ RY_1Y_2 \\ \oplus \\ RY_1^3 \end{matrix} \longrightarrow \begin{matrix} RY_2 \\ \oplus \\ RY_1^2 \end{matrix} \cdots \text{actually } Y_1^{(2)} \text{ here.} \longrightarrow RY_1 \longrightarrow R.$$

Then $\beta_0(k) = \varepsilon_0(R)$, $\beta_1(k) = \varepsilon_1(R)$, $\beta_2 = \varepsilon_1^2 + \binom{\varepsilon_1}{2}$, $\beta_3 = \varepsilon_1^3 + \varepsilon_1\varepsilon_2 + \binom{\varepsilon_1}{3}$.

↓ same Y_1 will repeat.

ii) Use Poincare's series:

- Def: Poincare's series of M is:

$$P_M^R(t) := \sum_{d=0}^{\infty} \beta_d(M) t^d.$$

- By ①, $R\langle Y\rangle$ min. free $\Rightarrow$ $k\langle Y\rangle$ has $\partial = 0$.

Then $\partial = 0$ means $k\langle Y\rangle = \bigotimes_{y \in Y} k\langle y\rangle$.

- Then we consider: if $|y|$ is odd,

$$\sum_{n=0}^{\infty} \dim_k(k\langle y\rangle_n) \cdot t^n = 1 + t^{2i-1}$$

↓ k ↓ only ky since $y^2 = 0$.

$|y|$ is even: $|y| = 2i$,

$$\sum_{n=0}^{\infty} \dim_k(k\langle y\rangle_n) \cdot t^n = \sum_{i=0}^{\infty} t^{(2i)i} = \frac{1}{1-t^{2i}}$$

↓ every degree

- In $k\langle Y\rangle = \bigotimes k\langle y\rangle$, tensoring is exactly poly. products.

Thus $P_k^R(t) = \prod_{y \in Y} \sum_{n=0}^{\infty} \dim_k(k\langle y\rangle_n) \cdot t^n$

$$= \frac{\prod_{i=1}^{\infty}(1+t^{2i-1})^{\varepsilon_{2i-1}(R)}}{\prod_{i=1}^{\infty}(1-t^{2i})^{\varepsilon_{2i}(R)}} \hookrightarrow |Y_{2i-1}| \text{ times since each degree has the same poly.}$$

③ Regularity in terms of $\varepsilon_i(R)$.

Lem: R is reg. $\Leftrightarrow \varepsilon_n(R)=0\ \forall n\geq 2 \Leftrightarrow \varepsilon_2(R)=0$.

Pf: • R reg. $\Rightarrow$ m gen. by reg. seq., $Kos(\underline{m})$ is the min. free

$\Rightarrow \varepsilon_n(R)=0$.

• $\varepsilon_2(R)=0 \Rightarrow$ no variables means $H_1(R\langle Y_1\rangle) = \underset{\| }{Kos(\underline{m})}_1 = 0$

$\Rightarrow$ m gen. by reg. seq.

3. Connection btw. $R\langle Y\rangle$ and $Q[X]$.

① Adjoining $[x]$ & $\langle y\rangle$.

✡ Lem: $A[x]\langle y : \partial(y)=x\rangle \longrightarrow A$ is a homotopy equiv. (of dga's).

Pf: Note here x is a variable with $\partial(x)=0$.

i) $|x|$ is even.

Then x is poly. var. and $|y|$ is odd hence exterior, $y^2=0$.

• The basis is: $\begin{cases} x^n,\ x^n y \\ \partial(x^n)=0 \text{ since } \partial(x)=0,\ \partial(x^n y)=x^{n+1}. \end{cases}$

• homotopy h is kind of "reversing" the ∂:

$$h(1)=h(x^n y)=0,\quad h(x^n)=x^{n-1}y.$$

Then $\begin{cases} (\partial h + h\partial)(x^n) = \partial(x^{n-1}y) = x^n \\ (\partial h + h\partial)(x^n y) = x^n y \end{cases}$ *: one of ∂h, $h\partial$ is 0, the other is recovering.

- Now for $i: A \hookrightarrow A[x]\langle y\rangle$,

$\pi : A[x]\langle y\rangle \to A$,

$i\pi = 0$ for x, y since $\pi(x) = \pi(y) = 0$.

Thus $\partial h + h\partial = id - i\pi$.

ii) $|x|$ odd, then x exterior and y divided power.

Then: $\begin{cases} \partial(y^{(n)}) = \partial(y)\, y^{(n-1)} = x y^{(n-1)}, \\ \partial(x y^{(n)}) = 0 + x^2 y^{(n-1)} = 0 \end{cases}$ → * well-behaved in any char.!!

Using similar h we have $\partial h + h\partial = id - i\pi$.

② KEY construction: acyclic closure of k over $k[X]$.

- First notice that $H_0(k[X]) = k$, $H_1(k[X]) = kX_1$

since $\partial(X_1) = 0$, $\partial(X_2) = 0$ by $\partial(k) \subset X^2$.

Similarly $H_n(k[X_{\geq n}]) = kX_n$.

- Let $Y_2 := \{y_x : x \in X_1\}$.

By ①, $k[X]\langle y_x : \partial(y_x) = x \in X_1\rangle \simeq k[X]/(X_1) \cong k[X_{\geq 2}]$.

This is "eliminating X_1 by adjoining Y_2".

- Similarly, by ②.ii),

$$k[X]\langle Y_2, Y_3\rangle \simeq k[X_{\geq 2}]\langle Y_3\rangle \simeq k[X_{\geq 3}].$$

$$k[X]\langle Y_{\leq n-1}, Y_n\rangle \simeq k[X_{\geq n-1}]\langle Y_n\rangle \simeq k[X_{\geq n}].$$

Thus $k[X]\langle Y_{\geq 2}\rangle \simeq k$ is $k[X]\langle Y\rangle$.

✡ Rmk: see 1.5.16 BEJ for $\partial(y_x)$ doesn't involve pure Y_n-terms in $k[X]\langle Y\rangle$, useful for their proofs in Gulliksen-Schoeller.

③ Thm: Fix $R\langle Y\rangle$ for k, $Q[X]$ for R. $\exists$ quasiisom. of loc'l dga:

$$R\langle Y_{\le n}\rangle \simeq k[X_{\le n}]$$

pf:

$\exists$ zigzag:

$$K^R \xleftarrow{\simeq} K^Q \otimes_Q Q[X] \xrightarrow{\simeq} k[X].$$

$K^Q \simeq k$, $K^Q \otimes_Q Q[X] \simeq R$

Then again by ②.ii),

$$K^R\langle Y_{\le n}\rangle \simeq k[X]\langle Y_{\le n}\rangle$$

$\simeq$ by definition! $R\langle Y_{\le n}\rangle$

$\simeq$ $k[X_{\le n}]$ by ②

since K^R is exactly $R\langle Y_1\rangle$.

④ Thm: $\varepsilon_i(R) = |Y_i| = |X_{i-1}|$.

pf:

Indeed, $|Y_i| = |H_{i-1}(R\langle Y_{\le i-1}\rangle)| = |H_{i-1}(k[X_{\le i-1}])| = |X_{i-1}|$

↓ by def. of adjoining Y_i

$= kX_{i-1}$

4. Criterion for CI.

① $\begin{cases} \varepsilon_1(R) = \text{embdim}(R) \\ \varepsilon_2(R) = \mu(I) \\ \varepsilon_3(R) = \mu(H_1(Kos(\underline{I}))) \text{ since } \varepsilon_3(R) = |X_2| = \text{no. of gen. of } Q[X_1] = Kos(\underline{I}). \end{cases}$

② Thm : R is C.I. $\Leftrightarrow \varepsilon_n(R) = 0 \ \forall n \geq 3$.

pf : $\begin{cases} R \text{ C.I.} \Rightarrow Q[X] = Q[X_1] \Rightarrow \varepsilon_3(R) = |X_2| = 0 \\ \varepsilon_n(R) = 0 \ \forall n \geq 3 \Rightarrow Q[X] \text{ finite} \Rightarrow R \text{ C.I.} \end{cases}$

③ ✡ Thm (Halperin) :

R C.I. $\Leftrightarrow \varepsilon_n(R) = 0$ for some n.

Rmk ✡ : if R not C.I., then $X_i \neq \emptyset$, $Y_i \neq \emptyset$ $\forall i \geq 1$.